battery will flow to all the electrical system (except starting system) of the vehicle. When the ignition switch is positioned in Starter (St), then the current will flow to the starter motor through the solenoid and thus the engine is started.



Figure 10.4.2 Ignition Switch

10.5 Difference between battery coil ignition system and magneto ignition system

	Battery coil ignition system	Magneto ignition system
1.	Electrical energy is supplied by the battery	Electrical energy is generated by the Magento
2.	Easy to start the engine	Little difficult to start the engine
3.	Difficult to start the engine, when the battery is low	The battery is not required; hence this problem will not arise
4.	Wiring circuit is complex	Wiring circuit is simple
5.	Spark intensity is good even at low engine speed.	Spark intensity is poor during low engine speed.
6.	More space is required	Less space is required
7.	Difficult in maintenance	Easy maintenance
8.	Does not work, if the battery fails	No need for battery
9.	Used in Cars, buses and trucks	Used in Moped, Scooter, Motor bikes.

10.6 Distributor

The Distributor does two functions. First, it has a set of contact breaker points, that works as a switch. When the CB points close, current flows through the coil. When the CB point open, current flow stops and coil produces a high tension current. A condenser is connected in parallel to the CB points. The condenser aids in the collapse of the magnetic field and help to reduce arching occurs between the CB points.

Second, the distributor is used to distribute the high tension current from the ignition coil to the spark plugs in the correct firing order. A coil wire delivers the high tension from the coil to the centre terminal of the distributor cap. Inside the cap, a rotor is on top of the distributor shaft. The distributor shaft is driven from the engine camshaft by a pair of spiral gears. When the rotor turns, the centre terminal of the distributor cap is connected by spark plug wires to the spark plugs. The spark plug wires carry the high tension current to the spark plug in the cylinder that is ready to fire.

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